

Ozone US-all Metric Inventory and Top-2 Summary

April 21, 2026

1 Source and reading guide

This note summarizes the following workbook:

```
D:/Github/spatial-skew-t/code/analysis/ozone/US-all/output/us-all/tables/comparison_top2.updated.xlsx
```

The goal is to keep the metric definitions, ranking logic, and top-2 model configurations in one manuscript-ready $\text{\LaTeX}$ file.

In the workbook, `rank.value` is always the quantity used to order the top-2 rows, whereas `score.value` is the raw metric itself. The detailed tables below collapse each `ranking_group` into one row with a winner and a runner-up. Because every top-2 entry in this workbook has `CMAQ=yes`, `CMAQ` is documented in the column glossary but omitted from the per-row configuration strings.

2 Metric inventory

3 High-level patterns

The global probabilistic-score block still leans toward Morris-baseline skew- t settings. In particular, CRPS is led by settings 35 and 11, and many Brier and Quantile-score winners at low-to-moderate quantile levels remain in the Morris lane.

Upper-tail scoring is more heterogeneous. AR2 wins Brier at $q = 0.95$ and $q = 0.995$, and it also takes Quantile-score wins at $q = 0.98, 0.99$, and 0.995 . MRTS wins Brier at $q = 0.9$, Quantile score at $q = 0.8$ and 0.9 , and the below-threshold Brier split at $q = 0.98$ and 0.99 .

Classification metrics show a precision-recall trade-off. High-threshold `Other` skew- t settings, especially settings 18 and 14 with `thr=85`, dominate precision and specificity. By contrast, MRTS appears most clearly in F1/Recall at $q = 0.98$, whereas calibration diagnostics remain mostly Morris or AR2: AR2 wins coverage at 0.8 and 0.9 , and Morris still dominates most PIT-based summaries.

Block	Metrics	Scenario dimension	Top-2 ranking rule
Probabilistic score	Brier, Quantile score, CRPS	Brier thresholds $q \in \{0, 0.1, \dots, 0.995\}$, quantile-score levels $q \in \{0.1, \dots, 0.995\}$, and one overall CRPS summary	<code>ranking_basis = rel_to_gaussian</code> ; smaller <code>rank_value</code> is better.
Tail-aware split score	Brier split	$q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ crossed with <code>above_threshold</code> , <code>all</code> , and <code>below_threshold</code>	<code>ranking_basis = rel_to_gaussian</code> ; smaller <code>rank_value</code> is better.
Classification	Accuracy, Precision, Recall, Specificity, F1	Event quantiles $q \in \{0.9, 0.95, 0.98, 0.99, 0.995\}$ with probability cutoff 0.5	<code>ranking_basis = raw_score</code> ; the workbook already sorts in the correct direction for each metric.
Calibration to target	Coverage, PIT mean, PIT variance	Coverage levels 0.8, 0.9, 0.95 plus overall PIT summaries	<code>ranking_basis = abs_gap_to_target</code> ; smaller absolute gap is better.
Raw uncertainty diagnostics	PIT KS, PIT uniformity MAE, PIT uniformity RMSE	Overall summaries	<code>ranking_basis = raw_score</code> ; lower values are better.

Table 1: Metric blocks represented in `comparison_top2.updated.xlsx`.

Workbook column(s)	Interpretation	How it appears in this draft
<code>metric_family</code> , <code>metric</code> <code>ranking_group</code>	Identify the metric block and the named metric. Source-side identifier for a specific metric/scenario combination.	Used as the sectioning logic and the first column of each detailed table. Suppressed in the detailed tables because <code>metric + scenario</code> is easier to read in the manuscript.
<code>target_level</code> , <code>threshold_value</code> , <code>target_value</code> <code>prediction_cutoff</code>	Define the nominal quantile/coverage level and the corresponding event or target. Probability cutoff used by classification metrics.	Condensed into the manuscript-side <i>Scenario</i> column. Included in the <i>Scenario</i> column for classification rows.
<code>band_type</code>	Split used by the Brier decomposition: <code>above_threshold</code> , <code>all</code> , or <code>below_threshold</code> .	Included in the <i>Scenario</i> column for Brier-split rows.
<code>ranking_basis</code>	Explains whether the workbook ranks by relative score, raw score, or absolute target gap.	Explained in the inventory table and used to interpret <code>rank_value</code> .
<code>rank_value</code>	Quantity actually used to assign rank 1 and rank 2.	Reported directly for both winner and runner-up.
<code>score_value</code> , <code>score_se</code> <code>setting</code>	Raw metric value and its standard error. Reproducible model identifier in the ozone US-all workflow.	Reported as $Score \pm SE$. Encoded inside each winner/runner-up label as S<setting> .
<code>method</code> , <code>knots</code> , <code>thresh</code> , <code>TS</code> , <code>ar2</code> , <code>mrts</code> <code>model_lane</code>	Core model-configuration fields. Broad model family: Gaussian reference, Morris baseline, AR2, MRTS, or another numeric variant.	Printed inside the winner/runner-up labels. Printed inside the winner/runner-up labels as Gaussian , Morris , AR2 , MRTS , or Other .
<code>CMAQ</code>	Indicator of whether CMAQ covariates are included.	Documented here only; all top-2 rows in this workbook have CMAQ=yes .

Table 2: Important workbook columns for manuscript use.

4 Detailed top-2 tables

Each winner/runner-up cell is a compact label containing the setting id, the model lane, the sampling family, and the key tuning fields `K`, `thr`, `TS`, `AR2`, and `MRTS`. The detailed tables retain the workbook’s top-2 ordering but replace the source-side `ranking_group` string with a more compact scenario label.

4.1 Probabilistic scores

For this block, `rank_value` is the relative score with respect to the Gaussian reference, so smaller values indicate stronger performance.

Table 3: Top-2 probabilistic-score summaries from
comparison_top2.updated.xlsx.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
Brier	q=0	S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	0.5834	0.00008342 $\pm$ 0.00008342	S104 / AR2 / skew-t (K=5, thr=0, TS=yes, AR2=yes, MRTS=-)	0.8289	0.0001185 $\pm$ 0.00007445
Brier	q=0.1	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9539	0.03689 $\pm$ 0.0009925	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.956	0.03697 $\pm$ 0.0009705
Brier	q=0.2	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9566	0.06051 $\pm$ 0.001701	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.9572	0.06055 $\pm$ 0.00148
Brier	q=0.3	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9629	0.07603 $\pm$ 0.001555	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9631	0.07604 $\pm$ 0.001287
Brier	q=0.4	S36 / Morris / skew-t (K=9, thr=0, TS=no, AR2=no, MRTS=-)	0.969	0.08652 $\pm$ 0.0006096	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9692	0.08653 $\pm$ 0.0007625
Brier	q=0.5	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9802	0.09271 $\pm$ 0.001121	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9807	0.09276 $\pm$ 0.001229
Brier	q=0.6	S34 / Morris / skew-t (K=7, thr=0, TS=no, AR2=no, MRTS=-)	0.988	0.09345 $\pm$ 0.0003049	S104 / AR2 / skew-t (K=5, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9884	0.09348 $\pm$ 0.0004491
Brier	q=0.7	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9849	0.08519 $\pm$ 0.0006925	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9859	0.08528 $\pm$ 0.0004477
Brier	q=0.8	S16 / Morris / skew-t (K=15, thr=50, TS=no, AR2=no, MRTS=-)	0.9868	0.0706 $\pm$ 0.001406	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9897	0.0708 $\pm$ 0.001739
Brier	q=0.9	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9916	0.04467 $\pm$ 0.00126	S51 / Morris / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=-)	0.9922	0.0447 $\pm$ 0.001301
Brier	q=0.95	S111 / AR2 / skew-t (K=7, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9856	0.02664 $\pm$ 0.001991	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9886	0.02672 $\pm$ 0.002042
Brier	q=0.98	S58 / Morris / skew-t (K=6, thr=50, TS=yes, AR2=no, MRTS=-)	0.9734	0.01257 $\pm$ 0.0007027	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9737	0.01257 $\pm$ 0.0006884
Brier	q=0.99	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9466	0.006876 $\pm$ 0.0006893	S8 / Morris / skew-t (K=5, thr=50, TS=no, AR2=no, MRTS=-)	0.95	0.0069 $\pm$ 0.0005906
Brier	q=0.995	S124 / AR2 / t (K=15, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9534	0.003842 $\pm$ 0.0001045	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9556	0.003851 $\pm$ 0.0001188
CRPS	overall	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.977	4.053 $\pm$ 0.01506	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9771	4.053 $\pm$ 0.01085
Quantile score	q=0.1	S72 / Morris / skew-t (K=15, thr=0, TS=yes, AR2=no, MRTS=-)	0.9927	4.366 $\pm$ 0.01051	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9927	4.366 $\pm$ 0.01304
Quantile score	q=0.2	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9959	7.223 $\pm$ 0.01741	S122 / AR2 / skew-t (K=15, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9959	7.223 $\pm$ 0.02098
Quantile score	q=0.3	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9975	9.185 $\pm$ 0.009203	S11 / Morris / skew-t (K=10, thr=0, TS=no, AR2=no, MRTS=-)	0.9976	9.185 $\pm$ 0.01037
Quantile score	q=0.4	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9982	10.4 $\pm$ 0.01359	S37 / Other / skew-t (K=11, thr=0, TS=no, AR2=no, MRTS=-)	0.9982	10.4 $\pm$ 0.01603
Quantile score	q=0.5	S37 / Other / skew-t (K=11, thr=0, TS=no, AR2=no, MRTS=-)	0.9983	10.9 $\pm$ 0.03013	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9984	10.9 $\pm$ 0.03219

Continued on next page

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
Quantile score	q=0.6	S60 / Morris / skew-t (K=7, thr=0, TS=yes, AR2=no, MRTS=-)	0.9988	10.69 $\pm$ 0.03775	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.999	10.7 $\pm$ 0.03958
Quantile score	q=0.7	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9988	9.74 $\pm$ 0.02604	S7 / Morris / skew-t (K=5, thr=0, TS=no, AR2=no, MRTS=-)	0.9992	9.743 $\pm$ 0.026
Quantile score	q=0.8	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9992	7.936 $\pm$ 0.02265	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9993	7.936 $\pm$ 0.0005303
Quantile score	q=0.9	S201 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=5)	0.9983	5.033 $\pm$ 0.0188	S7 / Morris / skew-t (K=5, thr=0, TS=no, AR2=no, MRTS=-)	0.9984	5.033 $\pm$ 0.0007798
Quantile score	q=0.95	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.9957	2.988 $\pm$ 0.009827	S119 / AR2 / skew-t (K=10, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9957	2.988 $\pm$ 0.004428
Quantile score	q=0.98	S119 / AR2 / skew-t (K=10, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9861	1.423 $\pm$ 0.01749	S36 / Morris / skew-t (K=9, thr=0, TS=no, AR2=no, MRTS=-)	0.9863	1.424 $\pm$ 0.02184
Quantile score	q=0.99	S108 / AR2 / skew-t (K=6, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9718	0.797 $\pm$ 0.01481	S67 / Morris / skew-t (K=9, thr=50, TS=yes, AR2=no, MRTS=-)	0.9732	0.7981 $\pm$ 0.01733
Quantile score	q=0.995	S108 / AR2 / skew-t (K=6, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9412	0.4413 $\pm$ 0.01264	S114 / AR2 / skew-t (K=8, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9428	0.442 $\pm$ 0.01541

4.2 Tail-aware Brier split

This table separates Brier performance into `above_threshold`, `all`, and `below_threshold` bands. As above, smaller `rank_value` indicates better relative performance to the Gaussian reference.

Table 4: Top-2 Brier-split summaries from `comparison.top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
∞	Brier split	q=0.9, band=above S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	0	0 $\pm$ 0	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.9543	0.2766 $\pm$ 0.001711
	Brier split	q=0.9, band=all S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9916	0.04467 $\pm$ 0.00126	S51 / Morris / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=-)	0.9922	0.0447 $\pm$ 0.001301
	Brier split	q=0.9, band=below S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.245	0.004428 $\pm$ 0.002788	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.44	0.007952 $\pm$ 0.004024
	Brier split	q=0.95, band=above S16 / Morris / skew-t (K=15, thr=50, TS=no, AR2=no, MRTS=-)	0.9317	0.3543 $\pm$ 0.01198	S38 / Morris / skew-t (K=6, thr=50, TS=no, AR2=no, MRTS=-)	0.9339	0.3551 $\pm$ 0.01365
	Brier split	q=0.95, band=all S111 / AR2 / skew-t (K=7, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9856	0.02664 $\pm$ 0.001991	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.9886	0.02672 $\pm$ 0.002042
	Brier split	q=0.95, band=below S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.4761	0.004014 $\pm$ 0.001834	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.6233	0.005256 $\pm$ 0.001358
	Brier split	q=0.98, band=above S71 / Morris / t (K=10, thr=75, TS=yes, AR2=no, MRTS=-)	0.8541	0.4612 $\pm$ 0.007674	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.8559	0.4621 $\pm$ 0.003531
	Brier split	q=0.98, band=all S58 / Morris / skew-t (K=6, thr=50, TS=yes, AR2=no, MRTS=-)	0.9734	0.01257 $\pm$ 0.0007027	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9737	0.01257 $\pm$ 0.0006884
	Brier split	q=0.98, band=below S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	0.9877	0.002187 $\pm$ 0.0002172	S204 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=10)	0.9971	0.002208 $\pm$ 0.0002081
	Brier split	q=0.99, band=above S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8009	0.5233 $\pm$ 0.04107	S71 / Morris / t (K=10, thr=75, TS=yes, AR2=no, MRTS=-)	0.8157	0.533 $\pm$ 0.03861
	Brier split	q=0.99, band=all S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9466	0.006876 $\pm$ 0.0006893	S8 / Morris / skew-t (K=5, thr=50, TS=no, AR2=no, MRTS=-)	0.95	0.0069 $\pm$ 0.0005906
	Brier split	q=0.99, band=below S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	0.9911	0.0007457 $\pm$ 0.00008788	S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	1	0.0007523 $\pm$ 0.00009287
	Brier split	q=0.995, band=above S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.736	0.5599 $\pm$ 0.06917	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.764	0.5812 $\pm$ 0.06674
	Brier split	q=0.995, band=all S124 / AR2 / t (K=15, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9534	0.003842 $\pm$ 0.0001045	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.9556	0.003851 $\pm$ 0.0001188
	Brier split	q=0.995, band=below S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	1	0.0002316 $\pm$ 0.000001534	S207 / MRTS / skew-t (K=1, thr=0, TS=yes, AR2=no, MRTS=15)	1.01	0.0002338 $\pm$ 0.000002013

4.3 Classification metrics

For this block, `rank_value` equals the raw classification metric after the workbook has applied the correct ranking direction.

Table 5: Top-2 classification summaries from `comparison_top2.updated.xlsx`.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
Accuracy	q=0.9, cutoff=0.5	S116 / AR2 / skew-t (K=9, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9383	0.9383 $\pm$ 0.001983	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.9383	0.9383 $\pm$ 0.002275
Accuracy	q=0.95, cutoff=0.5	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.9646	0.9646 $\pm$ 0.00339	S35 / Morris / skew-t (K=8, thr=0, TS=no, AR2=no, MRTS=-)	0.9645	0.9645 $\pm$ 0.003224
Accuracy	q=0.98, cutoff=0.5	S3 / Morris / skew-t (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.984	0.984 $\pm$ 0.00134	S120 / AR2 / skew-t (K=10, thr=50, TS=yes, AR2=yes, MRTS=-)	0.984	0.984 $\pm$ 0.001424
Accuracy	q=0.99, cutoff=0.5	S115 / AR2 / t (K=8, thr=75, TS=yes, AR2=yes, MRTS=-)	0.9918	0.9918 $\pm$ 0.0004617	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.9917	0.9917 $\pm$ 0.0008787
Accuracy	q=0.995, cutoff=0.5	S1 / Gaussian / gaussian (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.9956	0.9956 $\pm$ 0.000001658	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9956	0.9956 $\pm$ 0.00008504
F1	q=0.9, cutoff=0.5	S33 / Morris / skew-t (K=6, thr=0, TS=no, AR2=no, MRTS=-)	0.645	0.645 $\pm$ 0.005291	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.6442	0.6442 $\pm$ 0.002312
F1	q=0.95, cutoff=0.5	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.5744	0.5744 $\pm$ 0.01409	S55 / Morris / skew-t (K=5, thr=50, TS=yes, AR2=no, MRTS=-)	0.5728	0.5728 $\pm$ 0.02383
F1	q=0.98, cutoff=0.5	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.4652	0.4652 $\pm$ 0.004034	S65 / Morris / t (K=8, thr=75, TS=yes, AR2=no, MRTS=-)	0.4638	0.4638 $\pm$ 0.01459
F1	q=0.99, cutoff=0.5	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.4068	0.4068 $\pm$ 0.05383	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.3948	0.3948 $\pm$ 0.02491
F1	q=0.995, cutoff=0.5	S9 / Morris / t (K=5, thr=75, TS=no, AR2=no, MRTS=-)	0.3405	0.3405 $\pm$ 0.04048	S24 / Other / t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.3311	0.3311 $\pm$ 0.04538
Precision	q=0.9, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8409	0.8409 $\pm$ 0.03817	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.806	0.806 $\pm$ 0.06422
Precision	q=0.95, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.8028	0.8028 $\pm$ 0.04336	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.7658	0.7658 $\pm$ 0.02477
Precision	q=0.98, cutoff=0.5	S3 / Morris / skew-t (K=1, thr=0, TS=no, AR2=no, MRTS=-)	0.7555	0.7555 $\pm$ 0.05849	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.7446	0.7446 $\pm$ 0.05306
Precision	q=0.99, cutoff=0.5	S47 / Other / skew-t (K=6, thr=85, TS=no, AR2=no, MRTS=-)	0.7607	0.7607 $\pm$ 0.01071	S115 / AR2 / t (K=8, thr=75, TS=yes, AR2=yes, MRTS=-)	0.75	0.75 $\pm$ 0.03571
Precision	q=0.995, cutoff=0.5	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	0.9583	0.9583 $\pm$ 0.04167	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9545	0.9545 $\pm$ 0.04545
Recall	q=0.9, cutoff=0.5	S2 / Morris / max-stable (K=1, thr=75, TS=no, AR2=no, MRTS=-)	1	1 $\pm$ 0	S61 / Morris / skew-t (K=7, thr=50, TS=yes, AR2=no, MRTS=-)	0.5677	0.5677 $\pm$ 0.004929
Recall	q=0.95, cutoff=0.5	S4 / Morris / skew-t (K=1, thr=50, TS=no, AR2=no, MRTS=-)	0.4814	0.4814 $\pm$ 0.01243	S25 / Other / t (K=15, thr=75, TS=no, AR2=no, MRTS=-)	0.4765	0.4765 $\pm$ 0.01385
Recall	q=0.98, cutoff=0.5	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.3541	0.3541 $\pm$ 0.006249	S65 / Morris / t (K=8, thr=75, TS=yes, AR2=no, MRTS=-)	0.3505	0.3505 $\pm$ 0.01039
Recall	q=0.99, cutoff=0.5	S74 / Morris / t (K=15, thr=75, TS=yes, AR2=no, MRTS=-)	0.2853	0.2853 $\pm$ 0.03944	S203 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=5)	0.2764	0.2764 $\pm$ 0.01415
Recall	q=0.995, cutoff=0.5	S9 / Morris / t (K=5, thr=75, TS=no, AR2=no, MRTS=-)	0.2325	0.2325 $\pm$ 0.02558	S26 / Other / t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.2325	0.2325 $\pm$ 0.02558

Continued on next page

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
Specificity	q=0.9, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9962	0.9962 $\pm$ 0.002791	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.9922	0.9922 $\pm$ 0.005306
Specificity	q=0.95, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9973	0.9973 $\pm$ 0.001807	S14 / Other / skew-t (K=10, thr=85, TS=no, AR2=no, MRTS=-)	0.9955	0.9955 $\pm$ 0.001813
Specificity	q=0.98, cutoff=0.5	S18 / Other / skew-t (K=15, thr=85, TS=no, AR2=no, MRTS=-)	0.9981	0.9981 $\pm$ 0.0008954	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.998	0.998 $\pm$ 0.0005126
Specificity	q=0.99, cutoff=0.5	S57 / Morris / skew-t (K=6, thr=0, TS=yes, AR2=no, MRTS=-)	0.9995	0.9995 $\pm$ 0.00004179	S110 / AR2 / skew-t (K=7, thr=0, TS=yes, AR2=yes, MRTS=-)	0.9994	0.9994 $\pm$ 0.0001681
Specificity	q=0.995, cutoff=0.5	S15 / Morris / skew-t (K=15, thr=0, TS=no, AR2=no, MRTS=-)	1	1 $\pm$ 0.00004189	S107 / AR2 / skew-t (K=6, thr=0, TS=yes, AR2=yes, MRTS=-)	1	1 $\pm$ 0.00004189

4.4 Uncertainty and calibration

Coverage, PIT mean, and PIT variance are ranked by absolute gap to target; PIT KS, PIT MAE, and PIT RMSE are ranked directly on the raw score scale.

Table 6: Top-2 uncertainty and calibration summaries from
comparison_top2.updated.xlsx.

Metric	Scenario	Winner (setting/lane/model)	Rank value	Score $\pm$ SE	Runner-up (setting/lane/model)	Rank value	Score $\pm$ SE
Coverage	level=0.8	S117 / AR2 / skew-t (K=9, thr=50, TS=yes, AR2=yes, MRTS=-)	0.00008193	0.8001 $\pm$ 0.003869	S209 / MRTS / t (K=1, thr=75, TS=yes, AR2=no, MRTS=15)	0.001552	0.8016 $\pm$ 0.02369
Coverage	level=0.9	S113 / AR2 / skew-t (K=8, thr=0, TS=yes, AR2=yes, MRTS=-)	0.0002728	0.9003 $\pm$ 0.004716	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.0003933	0.8996 $\pm$ 0.007384
Coverage	level=0.95	S27 / Other / skew-t (K=2, thr=0, TS=no, AR2=no, MRTS=-)	0.00007324	0.9499 $\pm$ 0.0007276	S31 / Other / skew-t (K=4, thr=0, TS=no, AR2=no, MRTS=-)	0.0001149	0.9499 $\pm$ 0.0006859
PIT KS	overall	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.0229	0.0229 $\pm$ 0.001549	S59 / Morris / t (K=6, thr=75, TS=yes, AR2=no, MRTS=-)	0.03383	0.03383 $\pm$ 0.004204
PIT MAE	overall	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.005766	0.005766 $\pm$ 0.0009276	S117 / AR2 / skew-t (K=9, thr=50, TS=yes, AR2=yes, MRTS=-)	0.007056	0.007056 $\pm$ 0.0009916
PIT RMSE	overall	S68 / Morris / t (K=9, thr=75, TS=yes, AR2=no, MRTS=-)	0.006464	0.006464 $\pm$ 0.001116	S70 / Morris / skew-t (K=10, thr=50, TS=yes, AR2=no, MRTS=-)	0.00894	0.00894 $\pm$ 0.001535
PIT mean	overall	S7 / Morris / skew-t (K=5, thr=0, TS=no, AR2=no, MRTS=-)	0.0002274	0.5002 $\pm$ 0.02051	S54 / Morris / skew-t (K=5, thr=0, TS=yes, AR2=no, MRTS=-)	0.0003903	0.5004 $\pm$ 0.02017
PIT variance	overall	S73 / Morris / skew-t (K=15, thr=50, TS=yes, AR2=no, MRTS=-)	0.0002141	0.08355 $\pm$ 0.001212	S123 / AR2 / skew-t (K=15, thr=50, TS=yes, AR2=yes, MRTS=-)	0.000497	0.08383 $\pm$ 0.001293